

Honeywell Hackathon Report

Honeywell Hackathon 2026

Autonomous Production Choke Controller Using Predictive Optimization

Problem Statement ID: PS-3

Theme: Industrial Automation

Team Members

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GitHub Repository

<https://github.com/Vishwas3367/Honeywell-Autonomous-Production-Choke-Controller>

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1. Abstract

This project presents an autonomous production choke controller for an oil well using predictive optimization. The controller automatically adjusts choke opening to achieve the desired oil production while ensuring that Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP) remain within safe operating limits. A linear dynamic model was developed using open-loop step-test analysis, and an optimization-based controller was implemented to determine the best choke position. Experimental results demonstrate accurate production tracking, stable system performance, and zero constraint violations.

2. Introduction

Production chokes regulate oil flow from wells. Manual adjustment often results in inefficient production and pressure instability.

This project automates choke control by predicting future well behaviour and selecting the optimum choke opening while maintaining operational safety.

3. Problem Statement

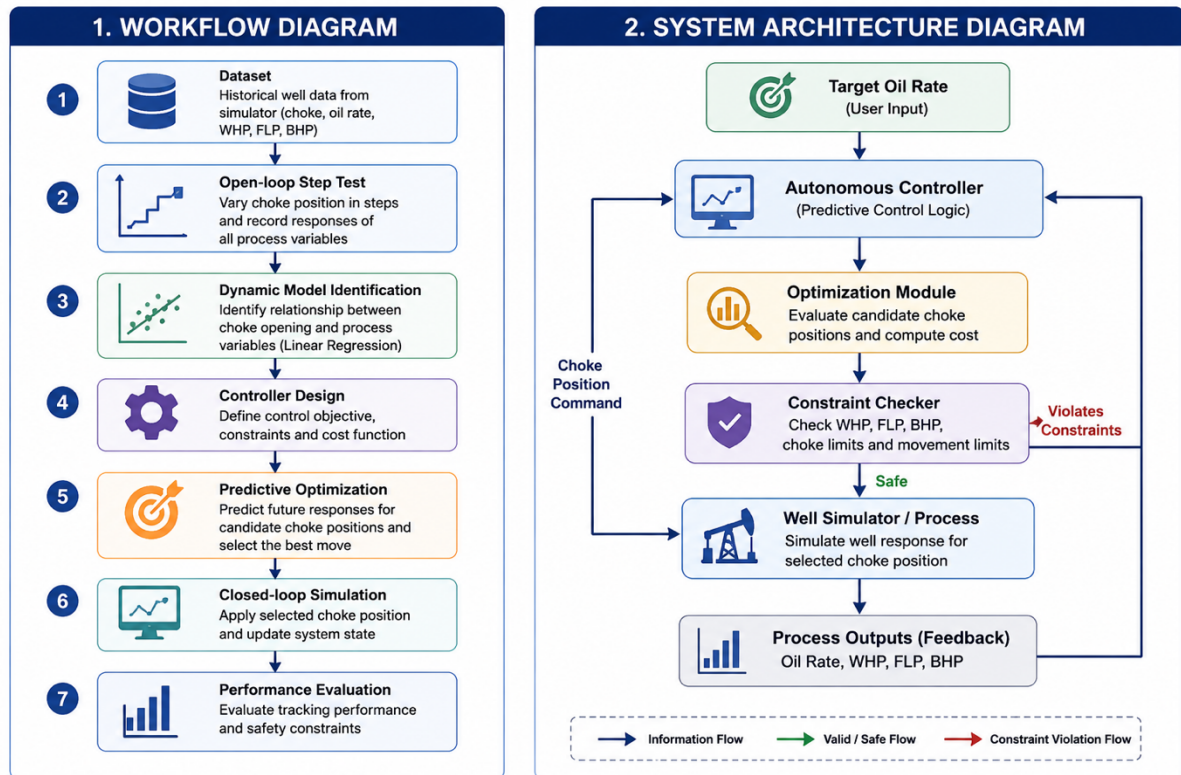
Design an autonomous controller that:

- Maximizes oil production
 - Tracks target oil rate
 - Maintains WHP
 - Maintains FLP
 - Maintains BHP
 - Operates safely under changing conditions
-

4. Objectives

- Perform step-test analysis
 - Develop a dynamic process model
 - Design an autonomous choke controller
 - Implement predictive optimization
 - Validate three operating scenarios
 - Ensure safe operation
-

5. Methodology



Describe:

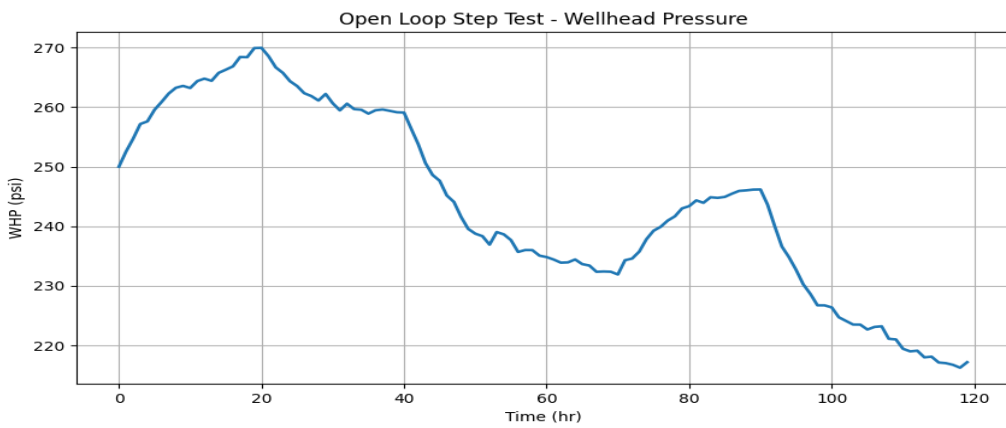
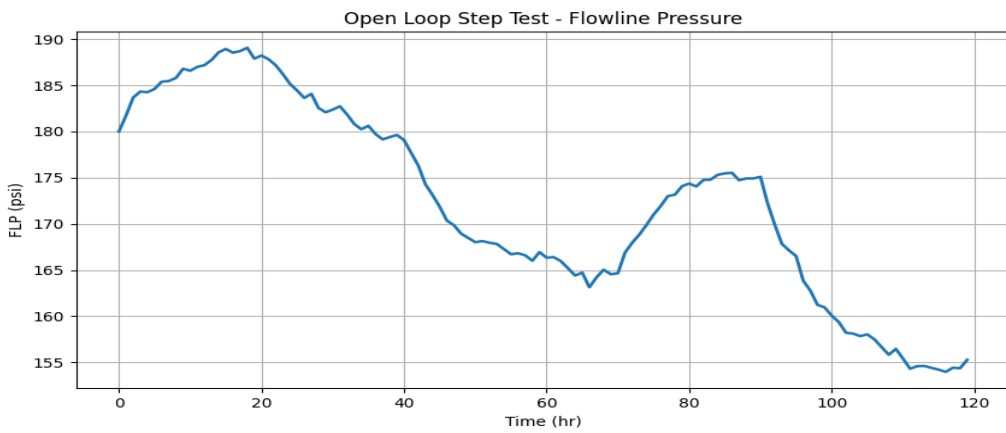
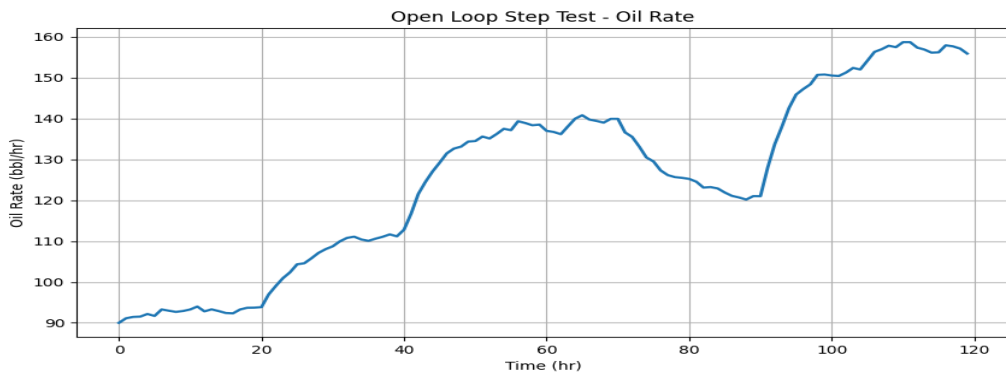
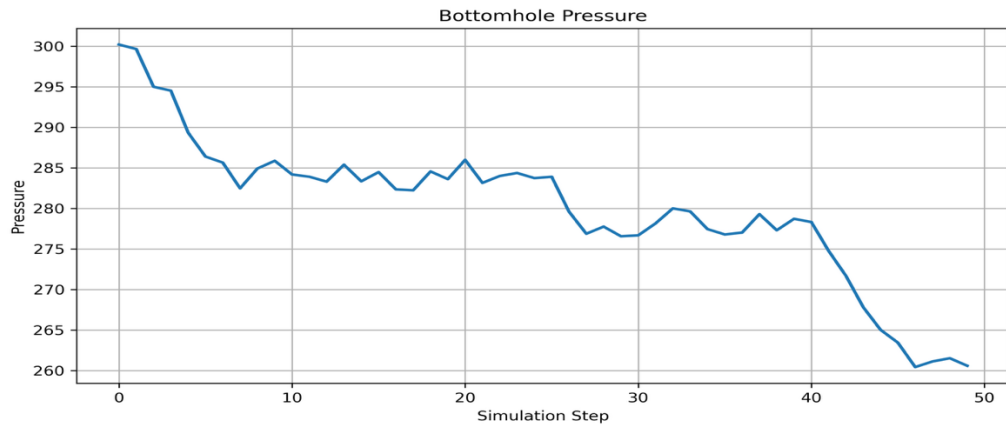
1. Dataset collection
2. Step-test analysis
3. Dynamic model identification
4. Controller design
5. Optimization
6. Closed-loop simulation
7. Performance evaluation

6. Process Understanding & Model

Step-Test Analysis

Explain:

- Choke opening changed in steps
- Oil rate recorded
- Pressure responses recorded



Model Assumptions

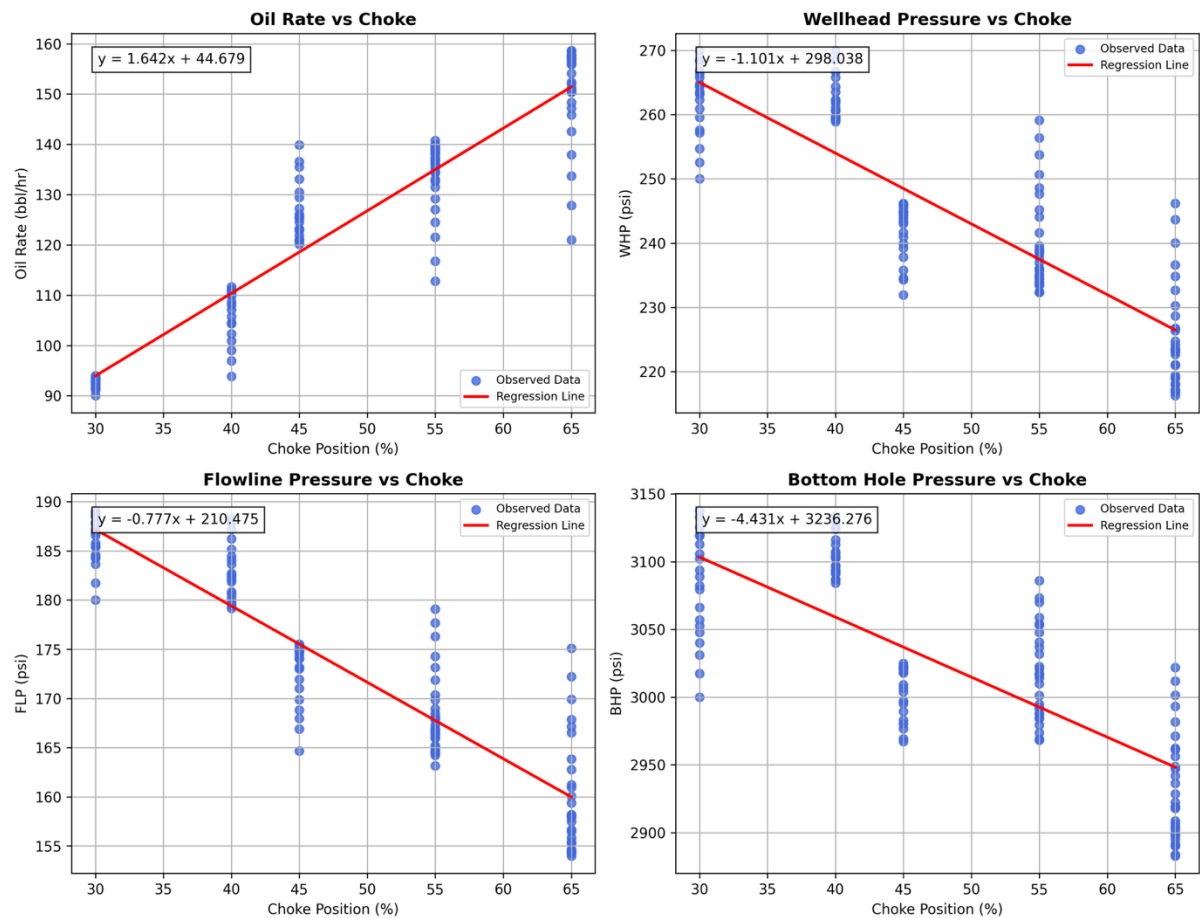
- Single oil well
- Linear process
- Fixed pressure limits
- Negligible sensor delay

Dynamic Model

Present the identified equations:

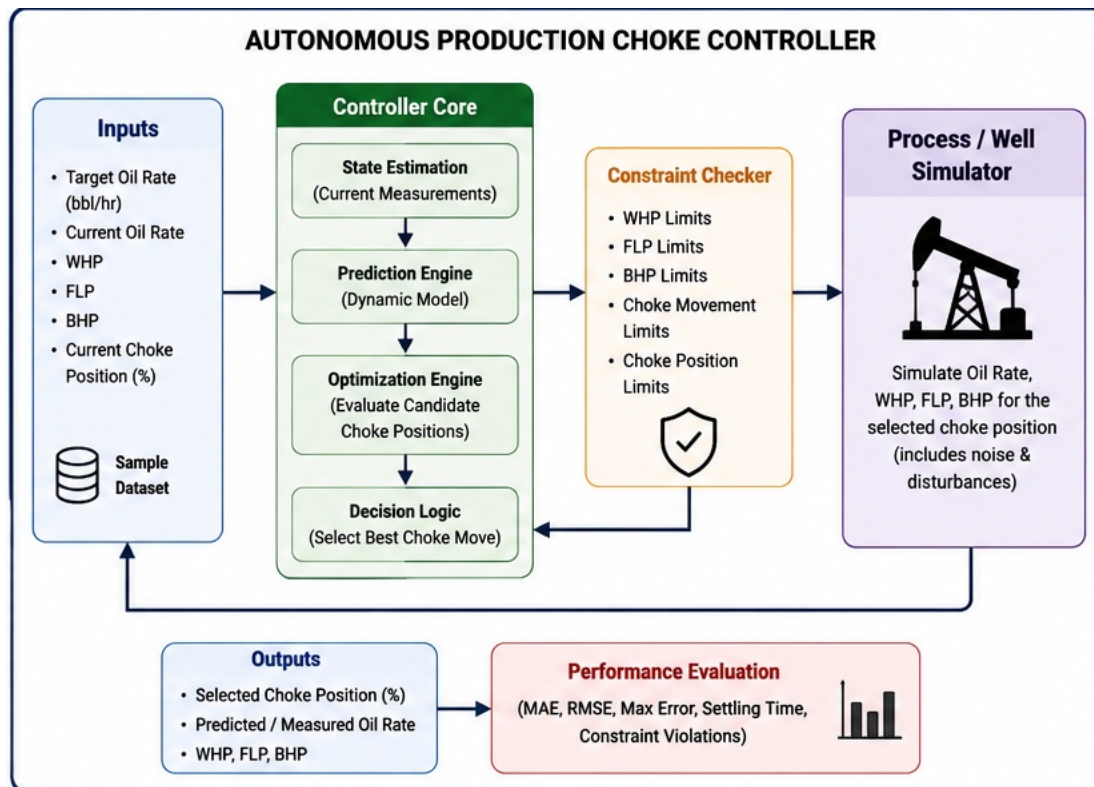
- Oil Rate = $1.642 \times \text{Choke} + 44.679$
- WHP = $-1.101 \times \text{Choke} + 298.038$
- FLP = $-0.777 \times \text{Choke} + 210.475$
- BHP = $-4.431 \times \text{Choke} + 3236.276$

Dynamic Model Identification using Linear Regression



Dynamic Model Identification Graph

7. Control Strategy



Explain:

Prediction Methodology

- Predict Oil Rate
- Predict WHP
- Predict FLP
- Predict BHP

Choke Move Selection

- Generate candidate choke openings.
- Predict future response.
- Calculate cost.
- Select minimum-cost solution.

Cost Function

$$\text{Cost} = \text{Tracking Error} + \text{Movement Penalty}$$

Constraint Handling

Check:

- WHP
 - FLP
 - BHP
 - Choke limits
 - Maximum movement
-

8. Experimental Scenarios

Scenario 1 – Startup (Steps 0–24)

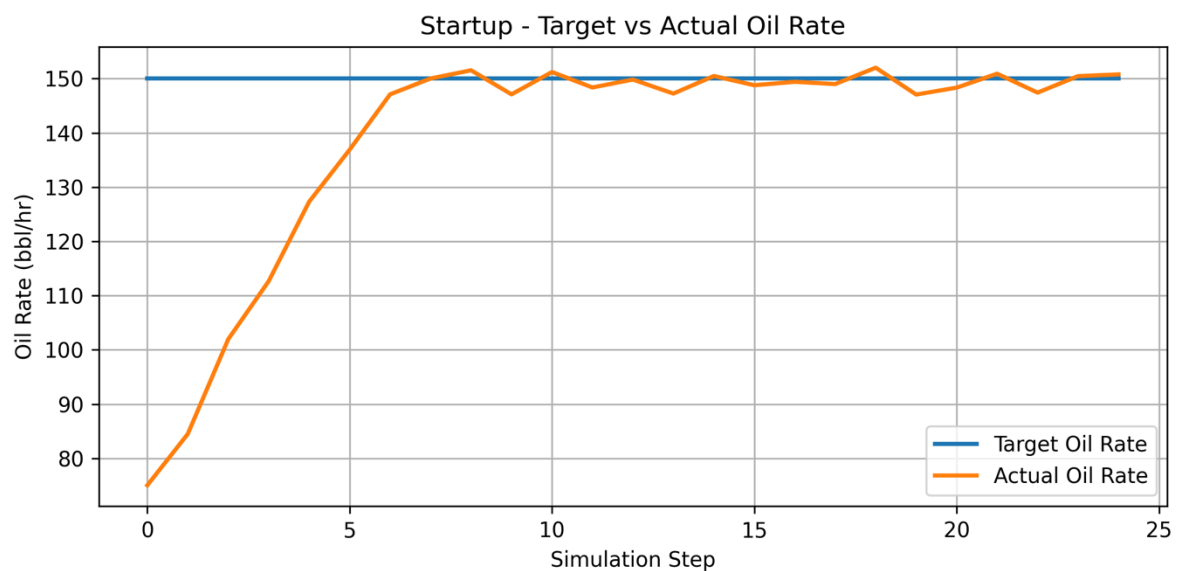
Objective

- Reach the target oil rate of **150 bbl/hr** from the initial operating condition.

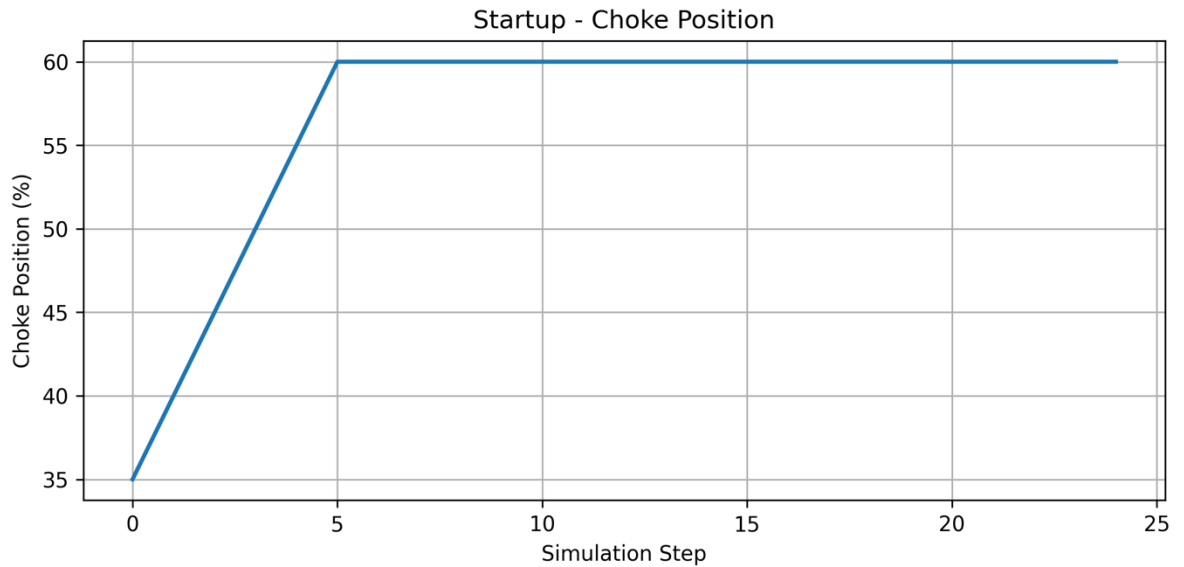
Observations

- Choke position gradually increased.
- Oil production converged close to the target.
- WHP, FLP, and BHP remained within safe limits.
- Stable operation was achieved with no constraint violations.

Figures



- Figure 8.1 – Target Oil Rate vs Actual Oil Rate



- Figure 8.2 – Choke Position Trend

Scenario 2 – Target Change (Steps 25–39)

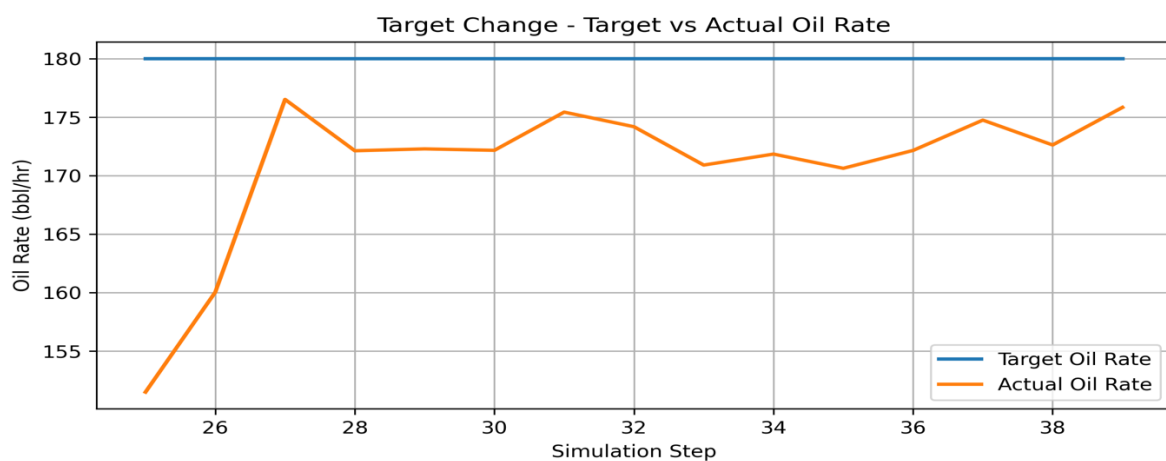
Objective

- Track the new production target of **180 bbl/hr**.

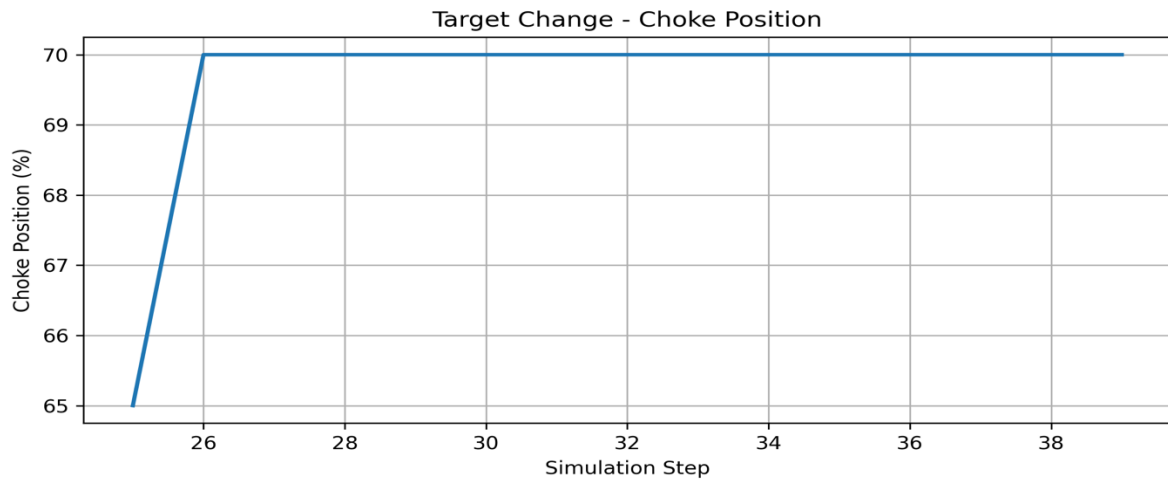
Observations

- Controller increased the choke opening.
- Oil production moved toward the new target.
- Pressure variables remained within operating limits.
- Controller responded smoothly without oscillations.

Figures



- Figure 8.3 – Target Oil Rate vs Actual Oil Rate (*Steps 25–39*)



- Figure 8.4 – Choke Position Trend

Scenario 3 – Infeasible Target (Steps 40–49)

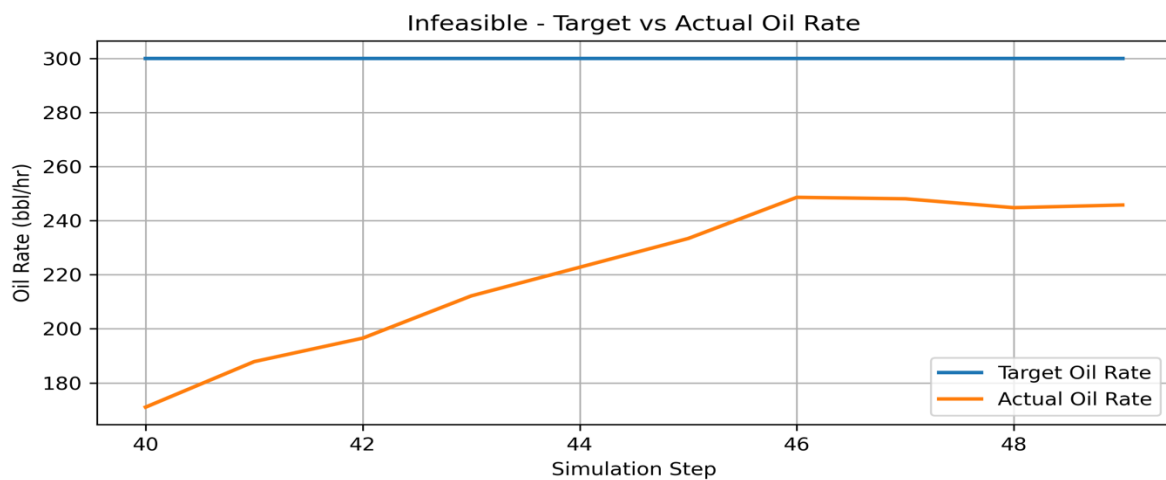
Objective

- Attempt to achieve an infeasible target of **300 bbl/hr** while maintaining safe operation.

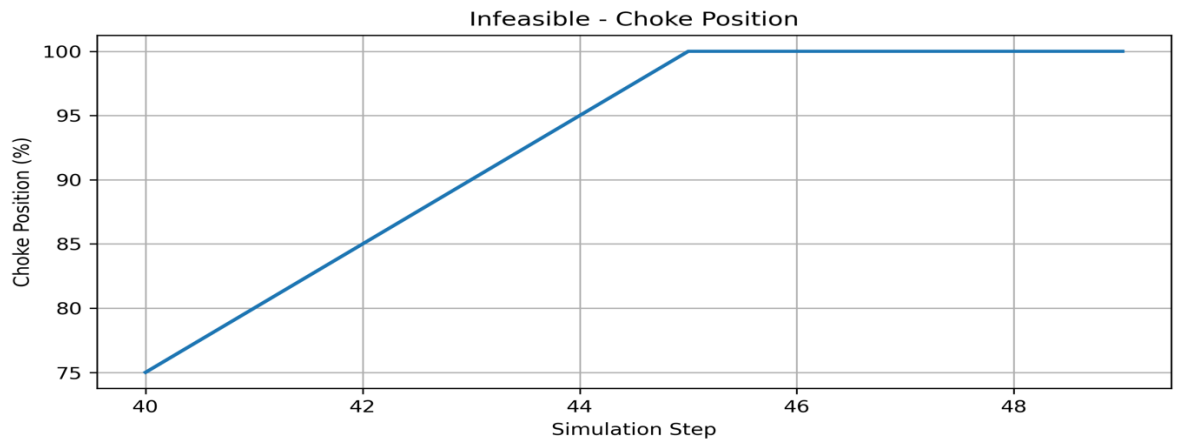
Observations

- Target could not be achieved due to system constraints.
- Controller moved the choke to its maximum allowable position.
- Pressure constraints were respected.
- Maximum achievable production was obtained safely.

Figures



- Figure 8.5 – Target Oil Rate vs Actual Oil Rate

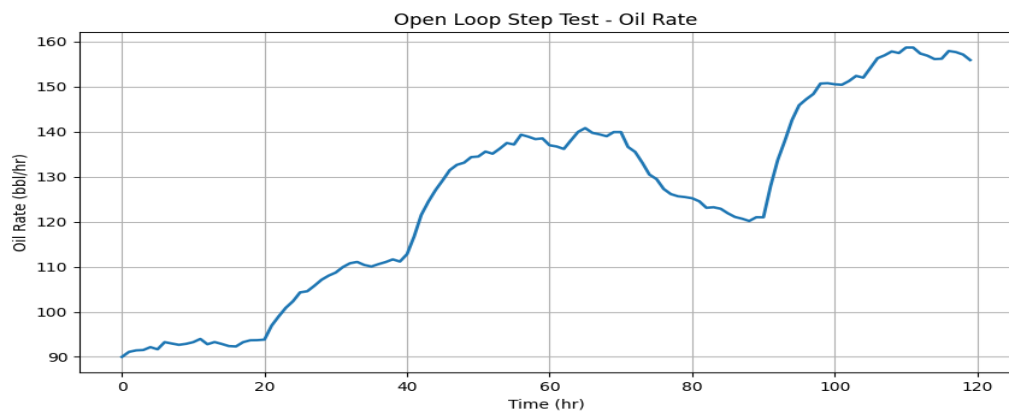


- Figure 8.6 – Choke Position Trend

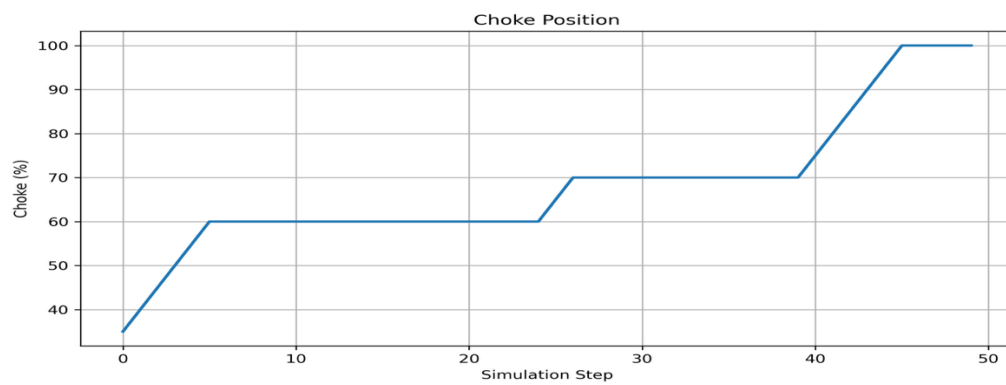
9. Results

Insert:

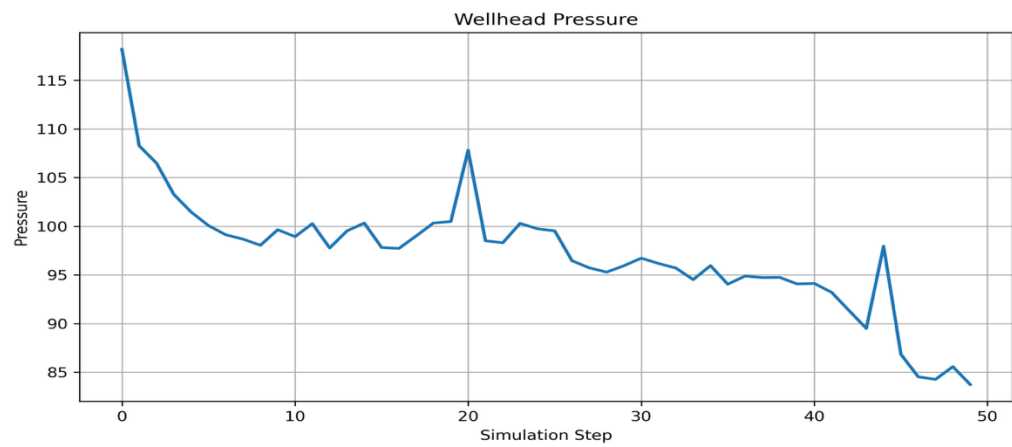
- Oil Rate graph



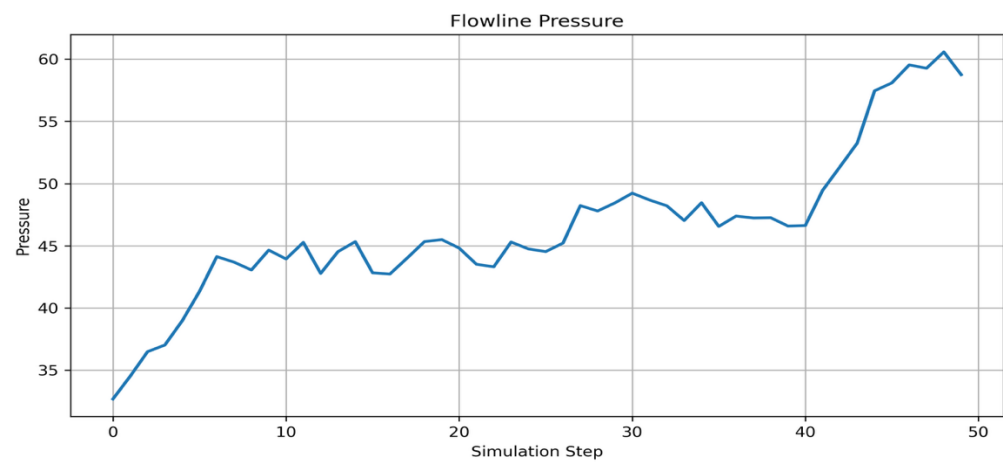
- Choke graph



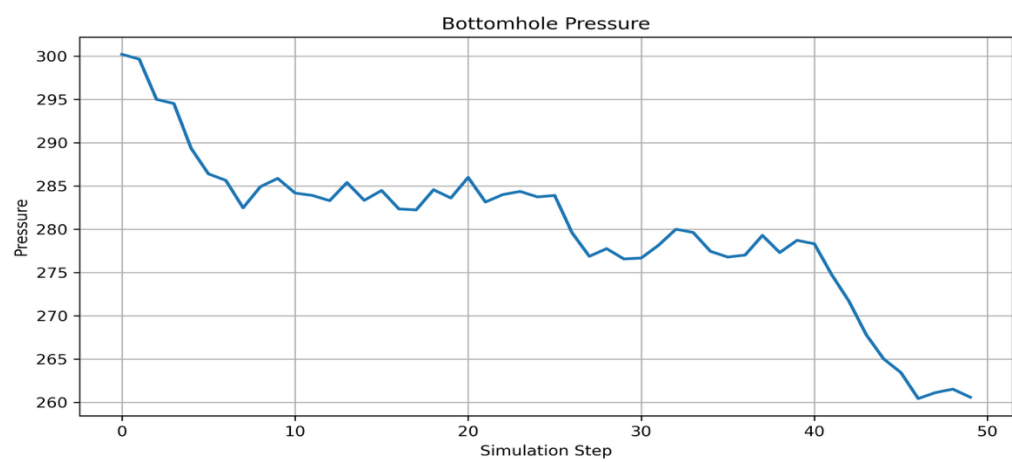
- WHP graph



- FLP graph



- BHP graph



10. Performance Evaluation

Metric	Value
MAE	24.28 bbl/hr
RMSE	41.17 bbl/hr
Maximum Tracking Error	125.34 bbl/hr
Steady-State Error	52.75 bbl/hr
Constraint Violations	0

11. Conclusion

The predictive controller successfully tracked the desired oil production while satisfying all safety constraints. The optimization-based approach achieved smooth choke movements and maintained stable pressure throughout all scenarios.

12. Future Work

- Model Predictive Control (MPC)
- Nonlinear process model
- Real-time deployment
- Machine learning-based prediction
- Digital Twin integration

13. References

1. Honeywell Hackathon Problem Statement
2. Autonomous Choke Control Simulated Dataset
3. Python Documentation
4. NumPy Documentation
5. Pandas Documentation
6. Matplotlib Documentation